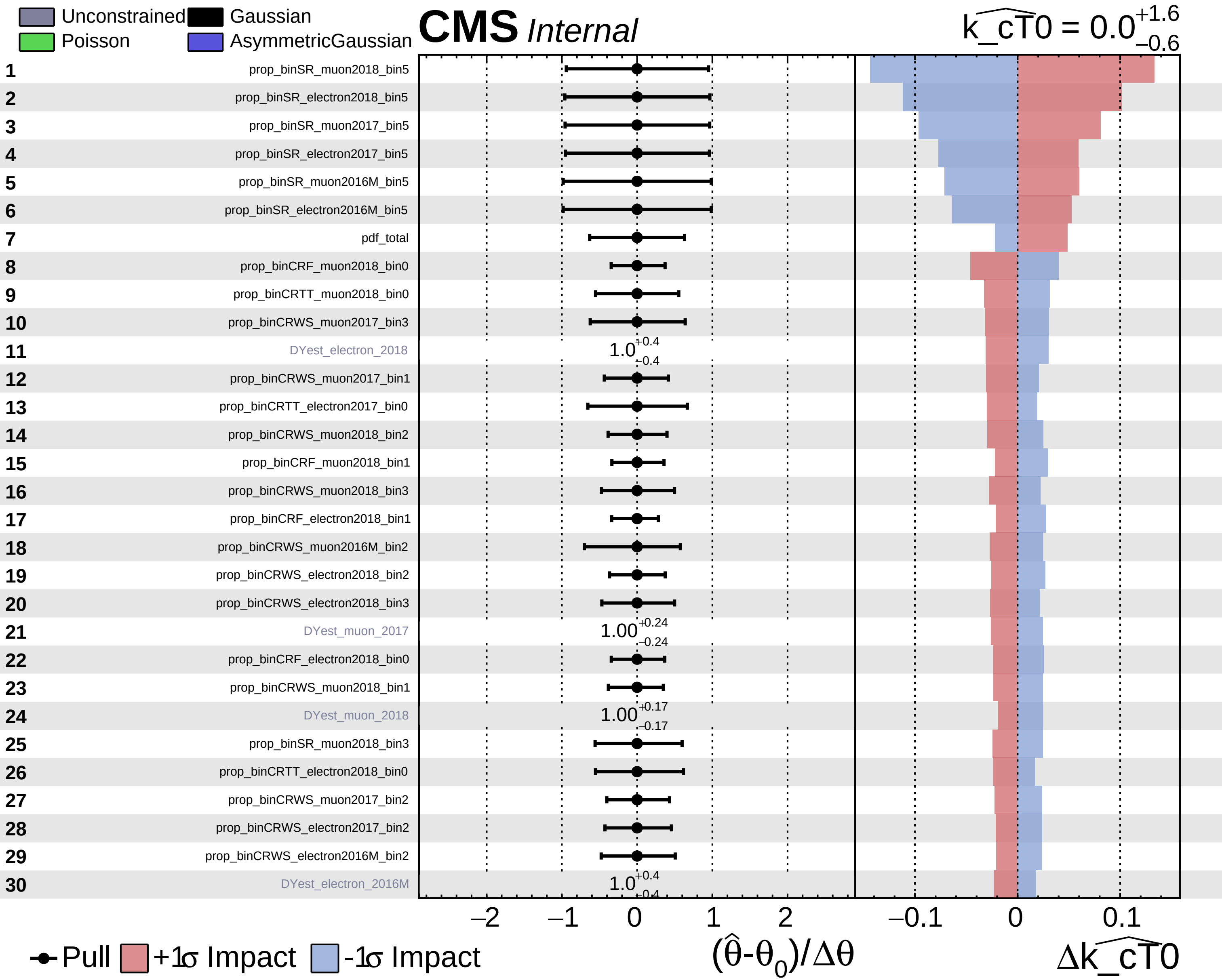
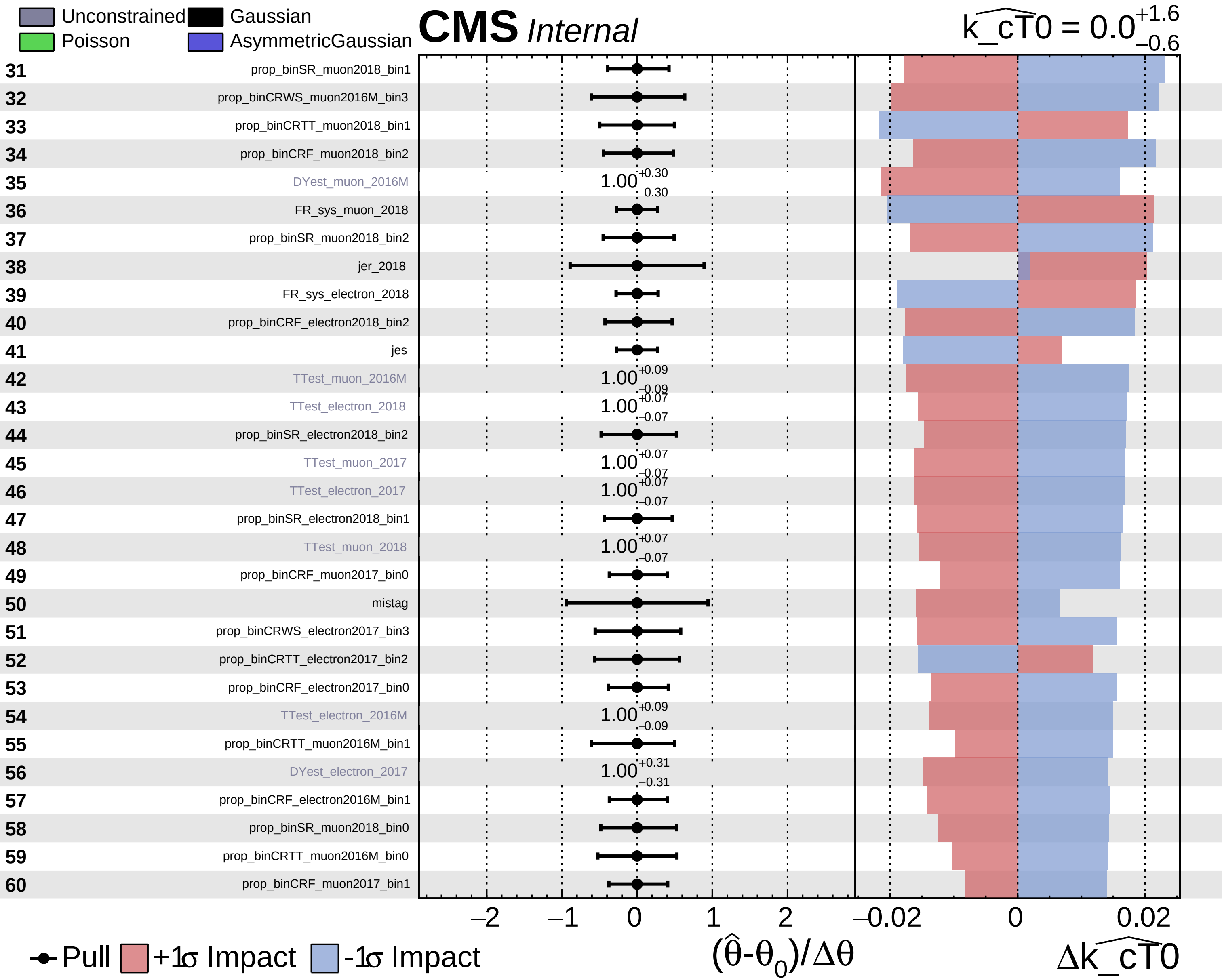
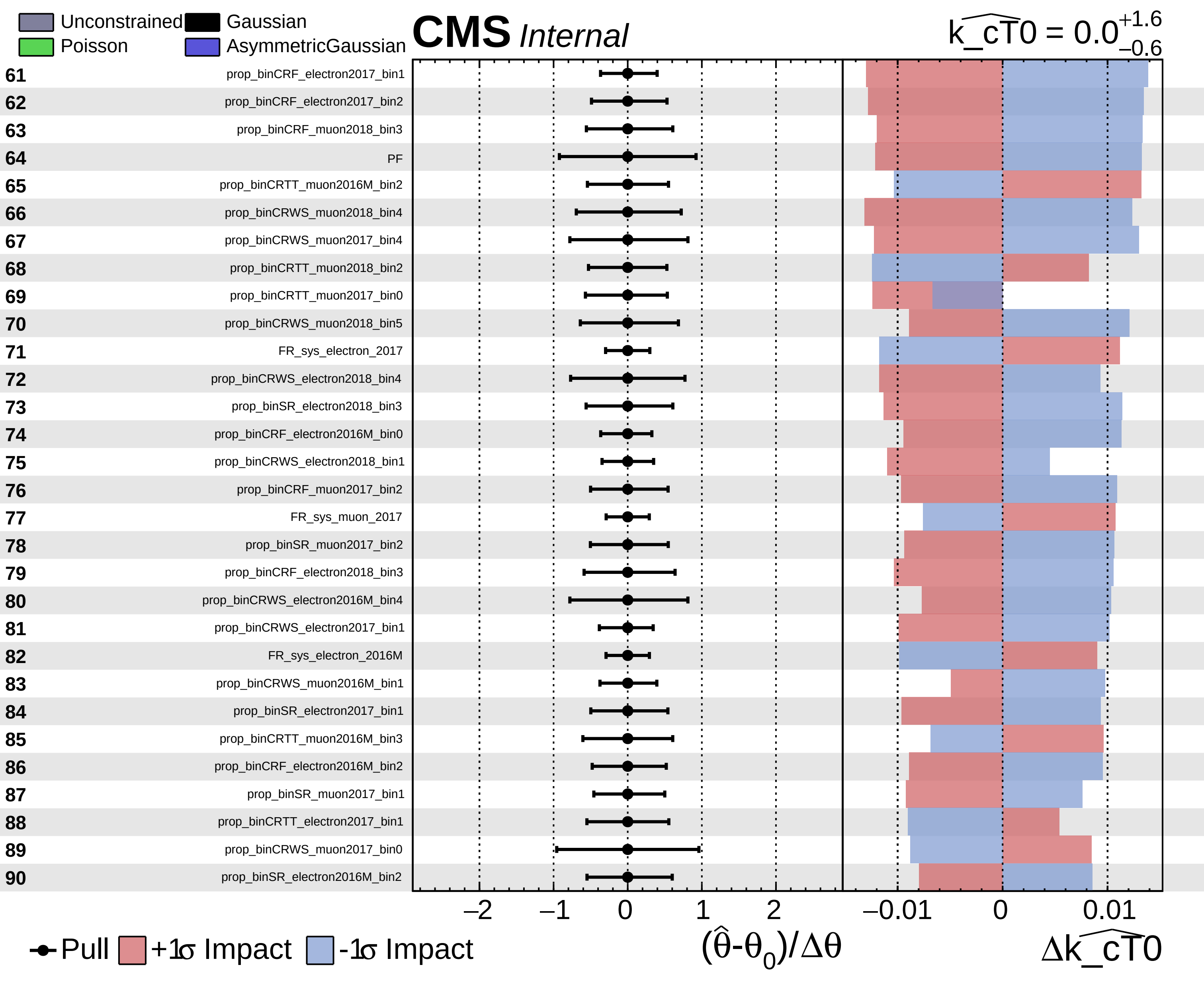


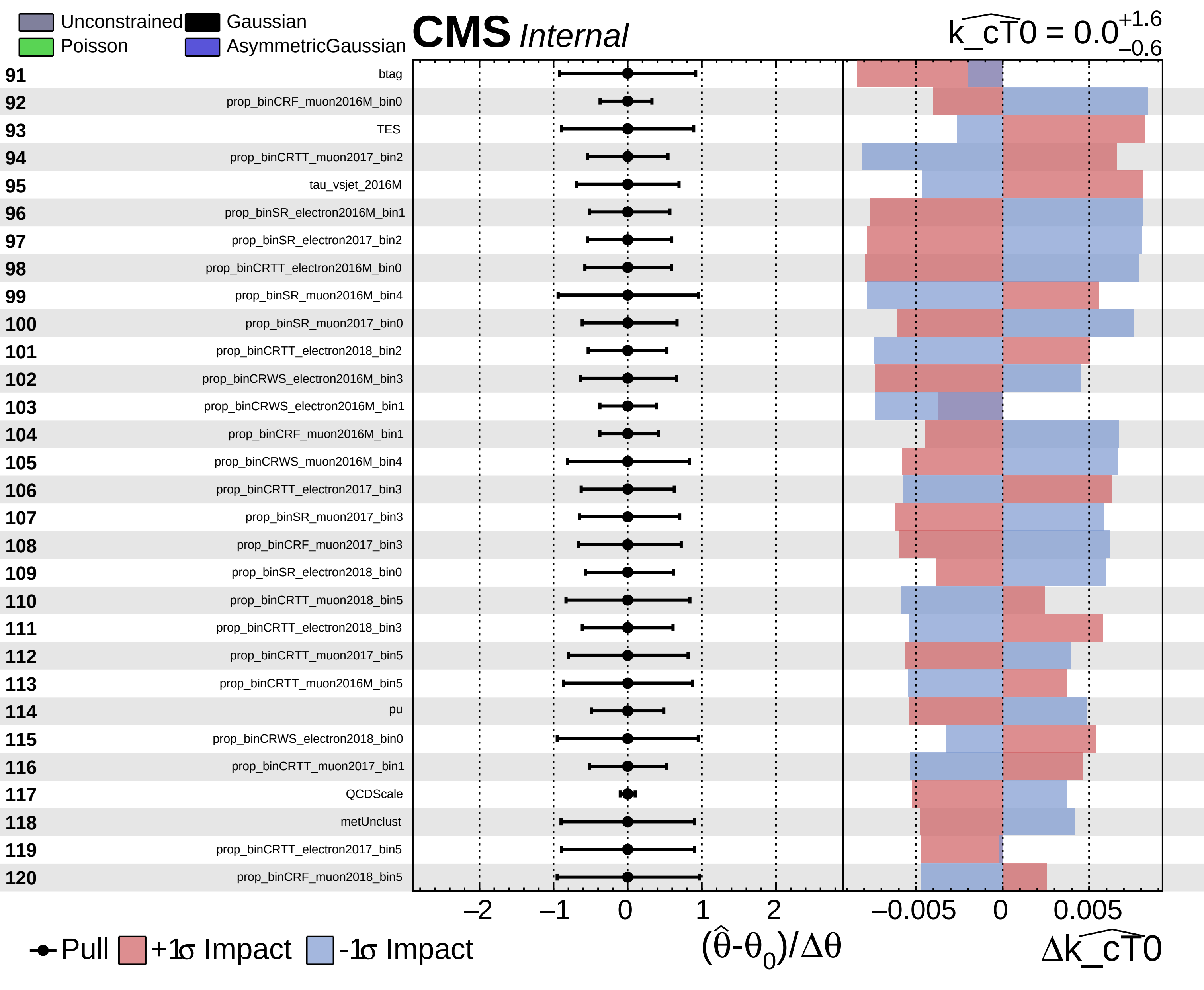
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = 0.0 +1.6 −0.6
1 prop_binSR_muon2018_bin5
2 prop_binSR_electron2018_bin5
3 prop_binSR_muon2017_bin5
4 prop_binSR_electron2017_bin5
5 prop_binSR_muon2016M_bin5
6 prop_binSR_electron2016M_bin5
7 pdf_total
8 prop_binCRF_muon2018_bin0
9 prop_binCRTT_muon2018_bin0
10 prop_binCRWS_muon2017_bin3
11 DYest_electron_2018
1.0 +0.4 −0.4
12 prop_binCRWS_muon2017_bin1
13 prop_binCRTT_electron2017_bin0
14 prop_binCRWS_muon2018_bin2
15 prop_binCRF_muon2018_bin1
16 prop_binCRWS_muon2018_bin3
17 prop_binCRF_electron2018_bin1
18 prop_binCRWS_muon2016M_bin2
19 prop_binCRWS_electron2018_bin2
20 prop_binCRWS_electron2018_bin3
21 DYest_muon_2017
1.00 +0.24 −0.24
22 prop_binCRF_electron2018_bin0
23 prop_binCRWS_muon2018_bin1
24 DYest_muon_2018
1.00 +0.17 −0.17
25 prop_binSR_muon2018_bin3
26 prop_binCRTT_electron2018_bin0
27 prop_binCRWS_muon2017_bin2
28 prop_binCRWS_electron2017_bin2
29 prop_binCRWS_electron2016M_bin2
30 DYest_electron_2016M
1.0 +0.4 −0.4
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0



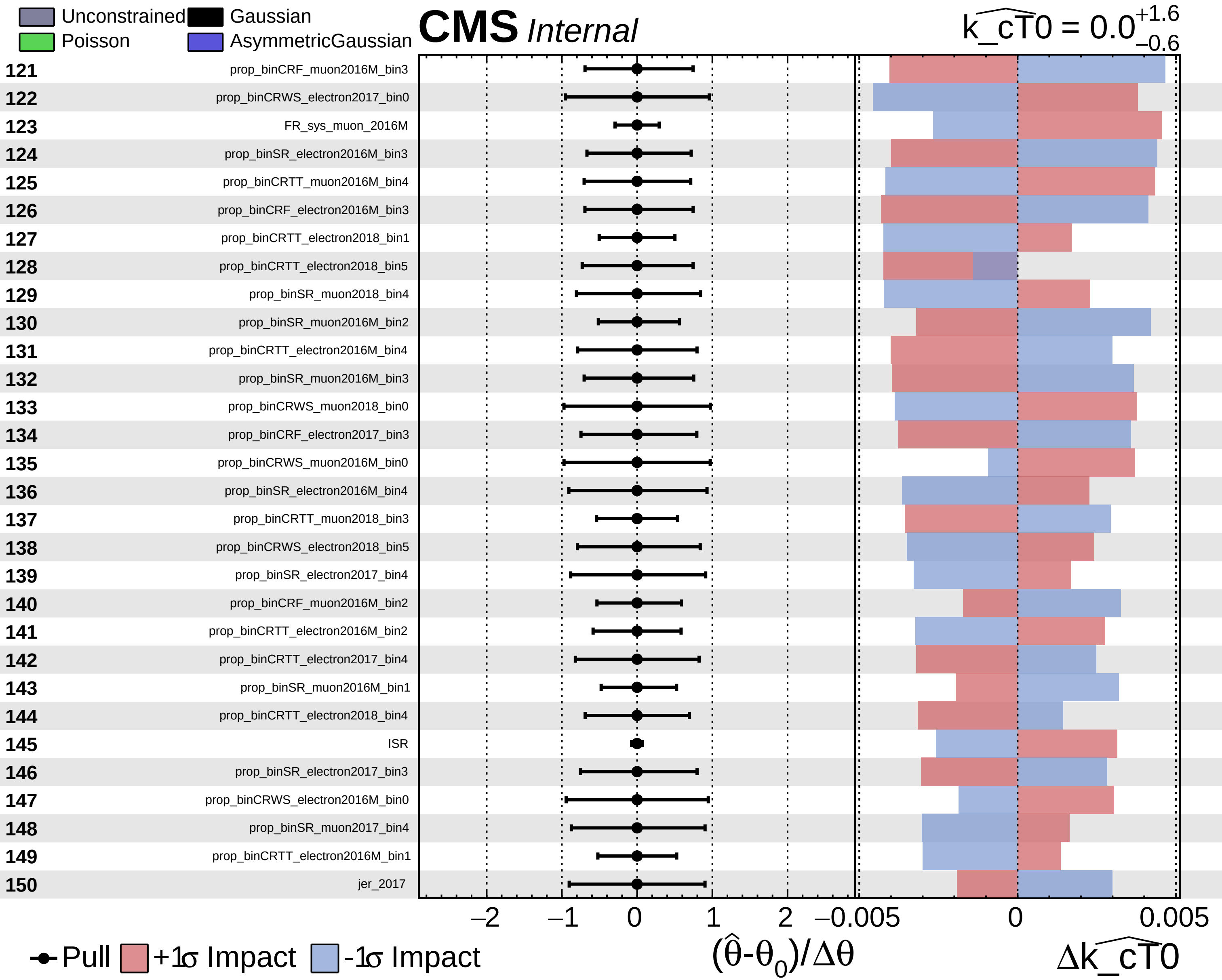
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = 0.0^{+1.6}_{-0.6}$
31 prop_binSR_muon2018_bin1
32 prop_binCRWS_muon2016M_bin3
33 prop_binCRTT_muon2018_bin1
34 prop_binCRF_muon2018_bin2
35 DYest_muon_2016M 1.00 +0.30 −0.30
36 FR_sys_muon_2018
37 prop_binSR_muon2018_bin2
38 jer_2018
39 FR_sys_electron_2018
40 prop_binCRF_electron2018_bin2
41 jes
42 TTest_muon_2016M 1.00 +0.09 −0.09
43 TTest_electron_2018 1.00 +0.07 −0.07
44 prop_binSR_electron2018_bin2
45 TTest_muon_2017 1.00 +0.07 −0.07
46 TTest_electron_2017 1.00 +0.07 −0.07
47 prop_binSR_electron2018_bin1
48 TTest_muon_2018 1.00 +0.07 −0.07
49 prop_binCRF_muon2017_bin0
50 mistag
51 prop_binCRWS_electron2017_bin3
52 prop_binCRTT_electron2017_bin2
53 prop_binCRF_electron2017_bin0
54 TTest_electron_2016M 1.00 +0.09 −0.09
55 prop_binCRTT_muon2016M_bin1
56 DYest_electron_2017 1.00 +0.31 −0.31
57 prop_binCRF_electron2016M_bin1
58 prop_binSR_muon2018_bin0
59 prop_binCRTT_muon2016M_bin0
60 prop_binCRF_muon2017_bin1
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



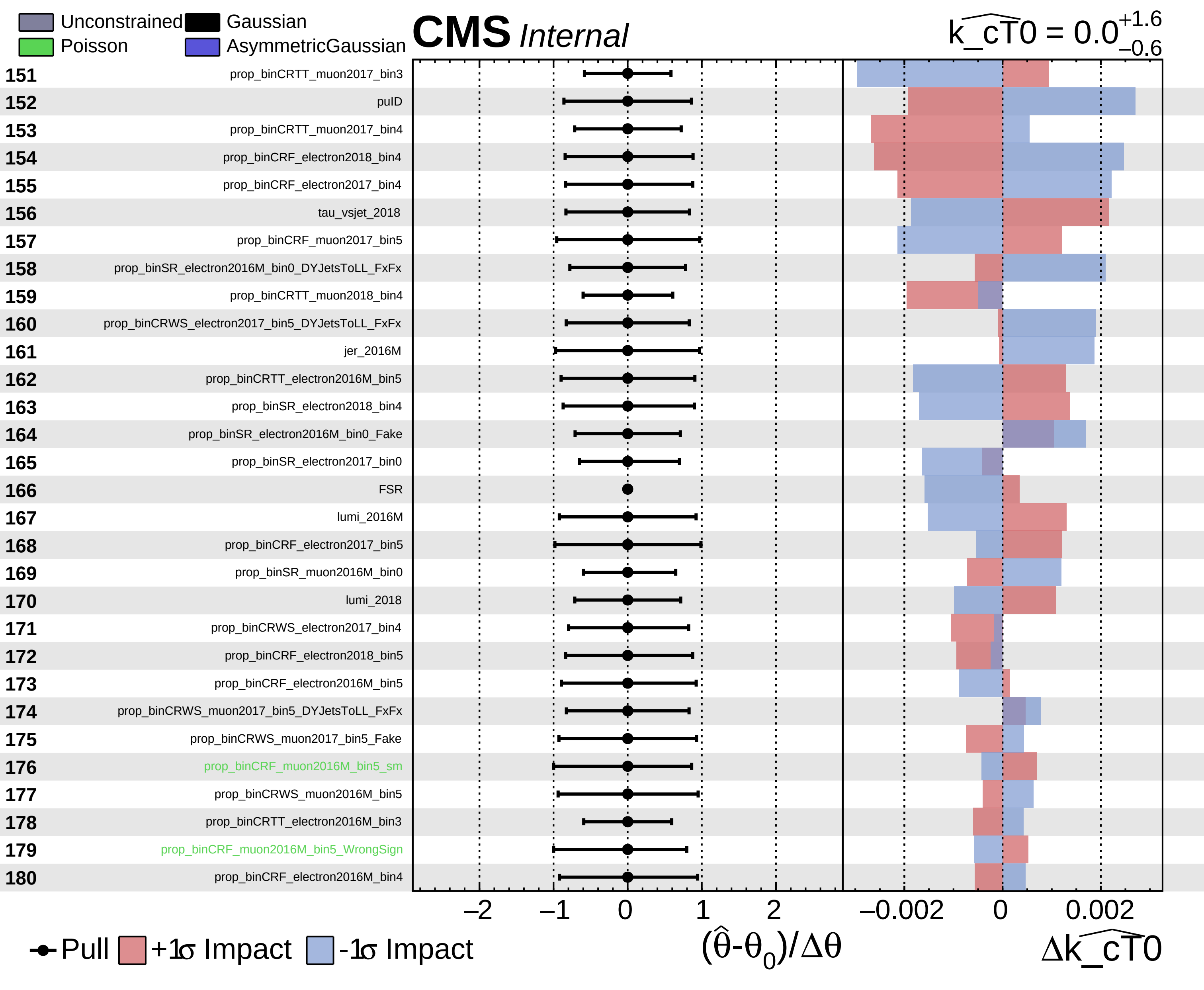
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = 0.0^{+1.6}_{-0.6}$
61 prop_binCRF_electron2017_bin1
62 prop_binCRF_electron2017_bin2
63 prop_binCRF_muon2018_bin3
64 PF
65 prop_binCRTT_muon2016M_bin2
66 prop_binCRWS_muon2018_bin4
67 prop_binCRWS_muon2017_bin4
68 prop_binCRTT_muon2018_bin2
69 prop_binCRTT_muon2017_bin0
70 prop_binCRWS_muon2018_bin5
71 FR_sys_electron_2017
72 prop_binCRWS_electron2018_bin4
73 prop_binSR_electron2018_bin3
74 prop_binCRF_electron2016M_bin0
75 prop_binCRWS_electron2018_bin1
76 prop_binCRF_muon2017_bin2
77 FR_sys_muon_2017
78 prop_binSR_muon2017_bin2
79 prop_binCRF_electron2018_bin3
80 prop_binCRWS_electron2016M_bin4
81 prop_binCRWS_electron2017_bin1
82 FR_sys_electron_2016M
83 prop_binCRWS_muon2016M_bin1
84 prop_binSR_electron2017_bin1
85 prop_binCRTT_muon2016M_bin3
86 prop_binCRF_electron2016M_bin2
87 prop_binSR_muon2017_bin1
88 prop_binCRTT_electron2017_bin1
89 prop_binCRWS_muon2017_bin0
90 prop_binSR_electron2016M_bin2
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



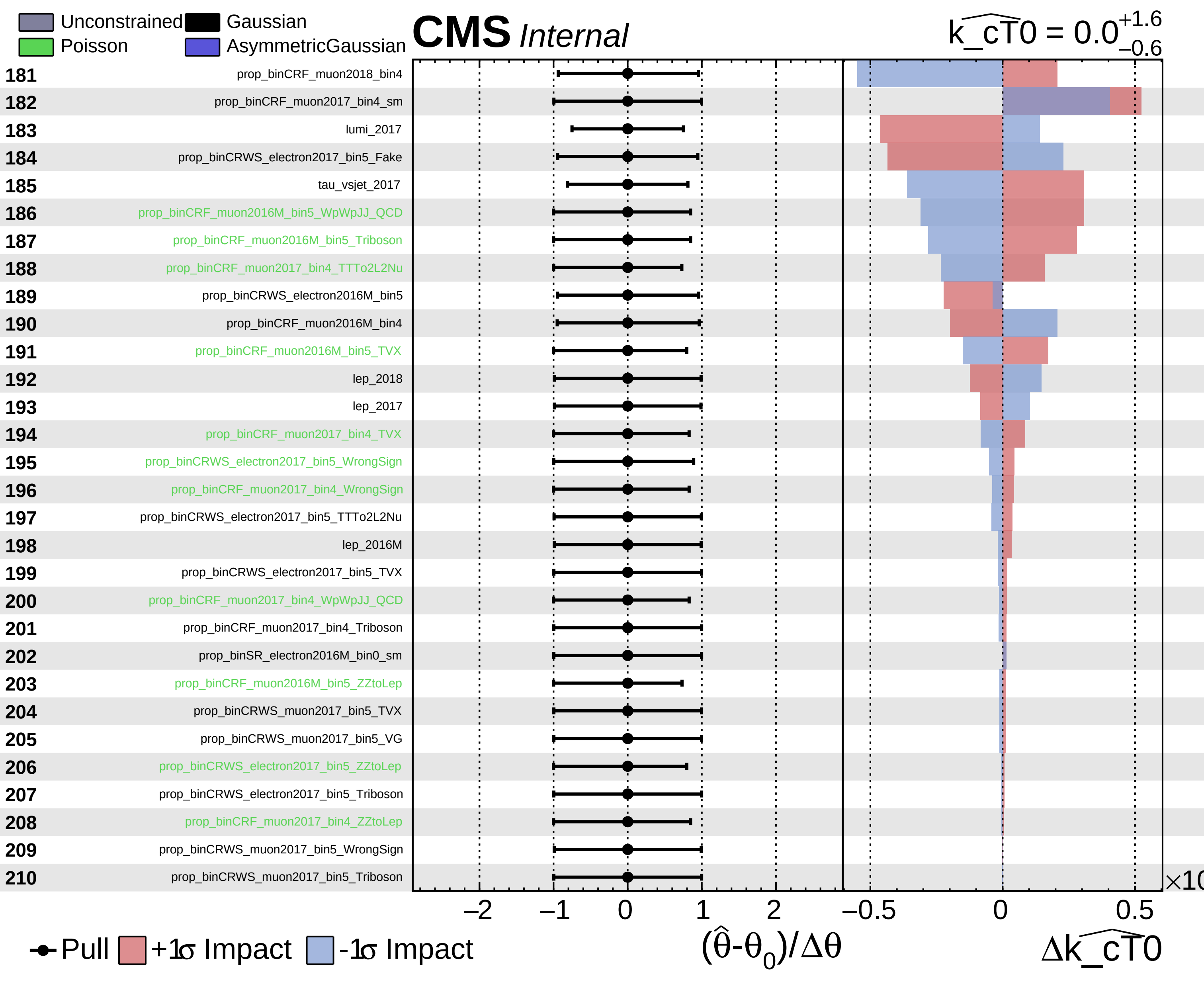
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = 0.0 +1.6 −0.6
91 btag
92 prop_binCRF_muon2016M_bin0
93 TES
94 prop_binCRTT_muon2017_bin2
95 tau_vsjet_2016M
96 prop_binSR_electron2016M_bin1
97 prop_binSR_electron2017_bin2
98 prop_binCRTT_electron2016M_bin0
99 prop_binSR_muon2016M_bin4
100 prop_binSR_muon2017_bin0
101 prop_binCRTT_electron2018_bin2
102 prop_binCRWS_electron2016M_bin3
103 prop_binCRWS_electron2016M_bin1
104 prop_binCRF_muon2016M_bin1
105 prop_binCRWS_muon2016M_bin4
106 prop_binCRTT_electron2017_bin3
107 prop_binSR_muon2017_bin3
108 prop_binCRF_muon2017_bin3
109 prop_binSR_electron2018_bin0
110 prop_binCRTT_muon2018_bin5
111 prop_binCRTT_electron2018_bin3
112 prop_binCRTT_muon2017_bin5
113 prop_binCRTT_muon2016M_bin5
114 pu
115 prop_binCRWS_electron2018_bin0
116 prop_binCRTT_muon2017_bin1
117 QCDScale
118 metUnclust
119 prop_binCRTT_electron2017_bin5
120 prop_binCRF_muon2018_bin5
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



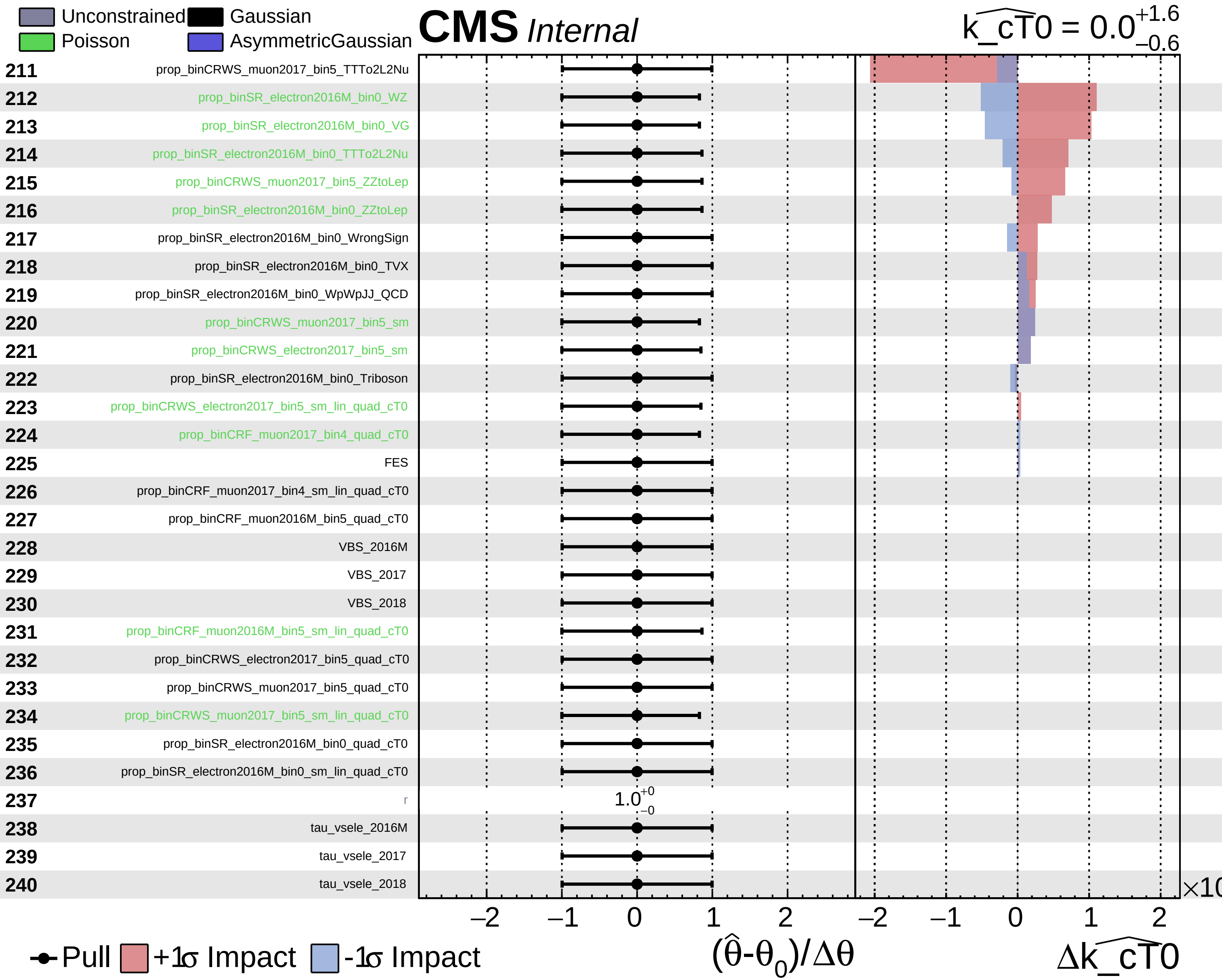
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = 0.0^{+1.6}_{-0.6}$
121 prop_binCRF_muon2016M_bin3
122 prop_binCRWS_electron2017_bin0
123 FR_sys_muon_2016M
124 prop_binSR_electron2016M_bin3
125 prop_binCRTT_muon2016M_bin4
126 prop_binCRF_electron2016M_bin3
127 prop_binCRTT_electron2018_bin1
128 prop_binCRTT_electron2018_bin5
129 prop_binSR_muon2018_bin4
130 prop_binSR_muon2016M_bin2
131 prop_binCRTT_electron2016M_bin4
132 prop_binSR_muon2016M_bin3
133 prop_binCRWS_muon2018_bin0
134 prop_binCRF_electron2017_bin3
135 prop_binCRWS_muon2016M_bin0
136 prop_binSR_electron2016M_bin4
137 prop_binCRTT_muon2018_bin3
138 prop_binCRWS_electron2018_bin5
139 prop_binSR_electron2017_bin4
140 prop_binCRF_muon2016M_bin2
141 prop_binCRTT_electron2016M_bin2
142 prop_binCRTT_electron2017_bin4
143 prop_binSR_muon2016M_bin1
144 prop_binCRTT_electron2018_bin4
145 ISR
146 prop_binSR_electron2017_bin3
147 prop_binCRWS_electron2016M_bin0
148 prop_binSR_muon2017_bin4
149 prop_binCRTT_electron2016M_bin1
150 jer_2017
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



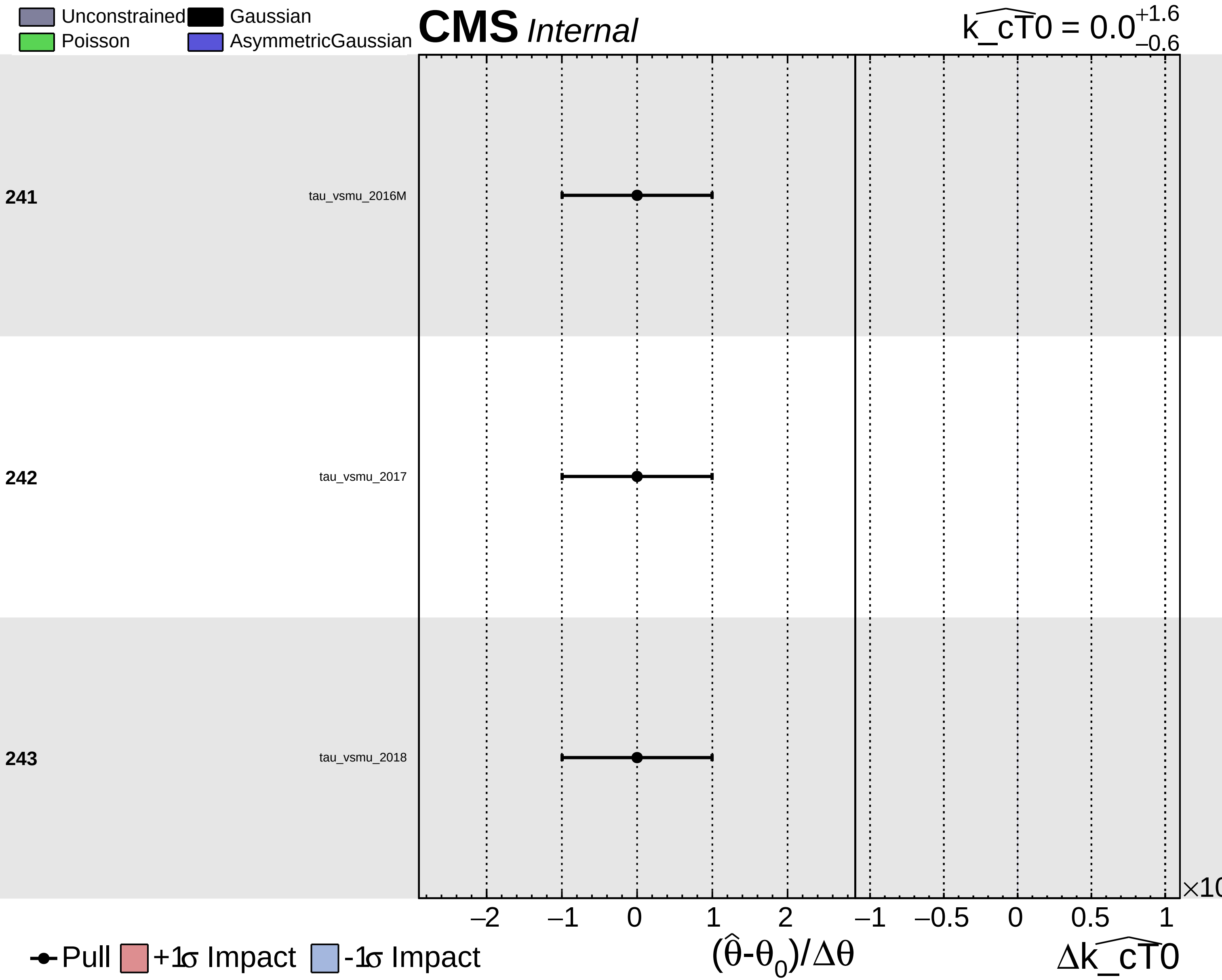
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = 0.0^{+1.6}_{-0.6}$
151 prop_binCRTT_muon2017_bin3
152 puID
153 prop_binCRTT_muon2017_bin4
154 prop_binCRF_electron2018_bin4
155 prop_binCRF_electron2017_bin4
156 tau_vsjet_2018
157 prop_binCRF_muon2017_bin5
158 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
159 prop_binCRTT_muon2018_bin4
160 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
161 jer_2016M
162 prop_binCRTT_electron2016M_bin5
163 prop_binSR_electron2018_bin4
164 prop_binSR_electron2016M_bin0_Fake
165 prop_binSR_electron2017_bin0
166 FSR
167 lumi_2016M
168 prop_binCRF_electron2017_bin5
169 prop_binSR_muon2016M_bin0
170 lumi_2018
171 prop_binCRWS_electron2017_bin4
172 prop_binCRF_electron2018_bin5
173 prop_binCRF_electron2016M_bin5
174 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
175 prop_binCRWS_muon2017_bin5_Fake
176 prop_binCRF_muon2016M_bin5_sm
177 prop_binCRWS_muon2016M_bin5
178 prop_binCRTT_electron2016M_bin3
179 prop_binCRF_muon2016M_bin5_WrongSign
180 prop_binCRF_electron2016M_bin4
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = 0.0^{+1.6}_{-0.6}$
181 prop_binCRF_muon2018_bin4
182 prop_binCRF_muon2017_bin4_sm
183 lumi_2017
184 prop_binCRWS_electron2017_bin5_Fake
185 tau_vsjet_2017
186 prop_binCRF_muon2016M_bin5_WpWpJJ_QCD
187 prop_binCRF_muon2016M_bin5_Triboson
188 prop_binCRF_muon2017_bin4_TTTo2L2Nu
189 prop_binCRWS_electron2016M_bin5
190 prop_binCRF_muon2016M_bin4
191 prop_binCRF_muon2016M_bin5_TVX
192 lep_2018
193 lep_2017
194 prop_binCRF_muon2017_bin4_TVX
195 prop_binCRWS_electron2017_bin5_WrongSign
196 prop_binCRF_muon2017_bin4_WrongSign
197 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
198 lep_2016M
199 prop_binCRWS_electron2017_bin5_TVX
200 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
201 prop_binCRF_muon2017_bin4_Triboson
202 prop_binSR_electron2016M_bin0_sm
203 prop_binCRF_muon2016M_bin5_ZZtoLep
204 prop_binCRWS_muon2017_bin5_TVX
205 prop_binCRWS_muon2017_bin5_VG
206 prop_binCRWS_electron2017_bin5_ZZtoLep
207 prop_binCRWS_electron2017_bin5_Triboson
208 prop_binCRF_muon2017_bin4_ZZtoLep
209 prop_binCRWS_muon2017_bin5_WrongSign
210 prop_binCRWS_muon2017_bin5_Triboson
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = 0.0 +1.6 −0.6
211 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
212 prop_binSR_electron2016M_bin0_WZ
213 prop_binSR_electron2016M_bin0_VG
214 prop_binSR_electron2016M_bin0_TTTo2L2Nu
215 prop_binCRWS_muon2017_bin5_ZZtoLep
216 prop_binSR_electron2016M_bin0_ZZtoLep
217 prop_binSR_electron2016M_bin0_WrongSign
218 prop_binSR_electron2016M_bin0_TVX
219 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
220 prop_binCRWS_muon2017_bin5_sm
221 prop_binCRWS_electron2017_bin5_sm
222 prop_binSR_electron2016M_bin0_Triboson
223 prop_binCRWS_electron2017_bin5_sm_lin_quad_cT0
224 prop_binCRF_muon2017_bin4_quad_cT0
225 FES
226 prop_binCRF_muon2017_bin4_sm_lin_quad_cT0
227 prop_binCRF_muon2016M_bin5_quad_cT0
228 VBS_2016M
229 VBS_2017
230 VBS_2018
231 prop_binCRF_muon2016M_bin5_sm_lin_quad_cT0
232 prop_binCRWS_electron2017_bin5_quad_cT0
233 prop_binCRWS_muon2017_bin5_quad_cT0
234 prop_binCRWS_muon2017_bin5_sm_lin_quad_cT0
235 prop_binSR_electron2016M_bin0_quad_cT0
236 prop_binSR_electron2016M_bin0_sm_lin_quad_cT0
237 r 1.0 +0 −0
238 tau_vsele_2016M
239 tau_vsele_2017
240 tau_vsele_2018
−2 −1 0 1 2
−2 −1 0 1 2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = 0.0 +1.6 −0.6
241
tau_vsmu_2016M
242
tau_vsmu_2017
243
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0